

A Transformer-based Forward Model For Aerosol and Ocean Data Retrieval from Polarimeters

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Abstract—This paper presents a Transformer-based deep learning model for retrieving aerosol and ocean optical parameters from polarimeter measurements. Traditional methods, e.g., Microphysical Aerosol Properties from Polarimetry (MAPP) algorithm, rely on coupled atmosphere–ocean vector radiative transfer (VRT) calculations, which are computationally expensive for large-scale data processing. To address this challenge, we develop a Transformer-based model to reproduce VRT outputs. The proposed model is trained and evaluated on a dataset of simulated VRT input-output pairs. Prediction performance is compared with convolutional neural networks (CNNs) and recurrent neural network (RNNs) trained on the same data. Experimental results show that the proposed Transformer-based model closely reproduces the VRT outputs, outperforming the CNN models and achieving accuracy comparable to the RNN models.

Index Terms—deep learning, transformers, polarimeter, ocean and aerosol retrieval.

I. INTRODUCTION

The Plankton, Aerosol, Cloud, and Ocean Ecosystem (PACE) mission is a recent NASA Earth-observing satellite designed to study the ocean and atmosphere system [1]. PACE carries multi-angle polarimeters that measure both the intensity and polarization of reflected sunlight across multiple wavelengths and viewing angles [2]. These polarimeter measurements provide rich information for estimating (“retrieving”) aerosol and ocean parameters, such as aerosol optical depth, particle size distribution, refractive index, single-scattering albedo, and ocean surface roughness [3], [4]

Conventional retrieval algorithms, such as the Microphysical Aerosol Properties from Polarimetry (MAPP), estimate the aerosol and ocean parameters through an iterative refining framework. In this framework, a state vector $\mathbf{x}$ describing aerosol and ocean parameters is iteratively updated by minimizing a cost function that balances the mismatch between observed and modeled polarimeter measurements and deviations from an a priori state [3]. A key step in this iterative refinement is to model polarimeter measurements using vector radiative transfer (VRT) calculations, which is a computationally intensive process, making the algorithm impractical for large-scale data retrieval [3], [5]

To reduce this computational burden, recent studies have investigated deep learning (DL) approaches for accelerating

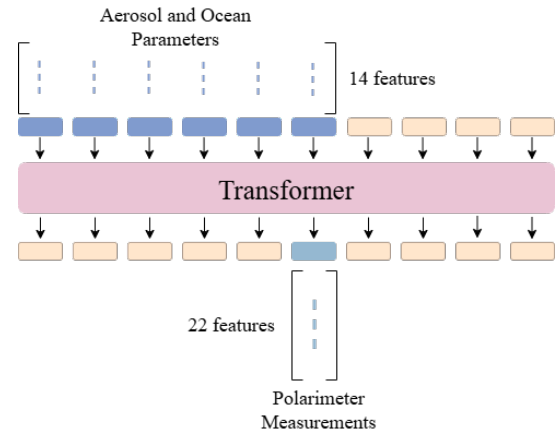


Fig. 1. The proposed Transformer-based method is designed to replicate the vector radiative transfer (VRT) used in the MAPP algorithm [6], to accelerate the retrieval time.

polarimeter retrievals [6]–[8]. In this approach, neural networks are trained on the dataset to learn a direct mapping of VRT inputs and outputs, i.e., from aerosol and ocean parameters to polarimeter measurements. Once trained, these models can replace VRT calculations to enable fast inference, therefore, accelerating the overall retrieval process.

Previous work has shown that various DL models, including feedforward neural networks (FNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs), can reproduce VRT outputs while greatly reducing computation time [7]. This paper presents a new DL model, i.e., a Transformer-based model, to replace the VRT calculations. Transformer models employ attention mechanisms to capture dependencies across multiple observations and offer a flexible architecture for modeling structured input data [9], [10]. To train this Transformer model, we use a dataset generated by pre-computing VRT calculations over a wide range of input state vectors. The results are stored as input-output samples, where each sample consists of a 14-dimensional input vector (aerosol and ocean parameters and sun instrument geometry parameters) and the corresponding 22-dimensional output vector (polarimeter measurements). In total, 2 million samples are used for model training, and an additional 200,000 samples

are used for testing. We apply a sliding window construction: input vectors are grouped into overlapping windows of length T and used as the model input, while a supervision target is associated with the final index of the window.

As illustrated in Fig. 1, the objective of this work is to replace the computationally intensive VRT calculations used in the MAPP algorithm with a Transformer model. The main contributions of this paper are as follows:

- We develop a Transformer model operating on sliding windows of aerosol and ocean parameters to predict the corresponding polarimeter measurements.
- We quantify the performance of this Transformer model with CNN and RNN models using the same dataset, loss functions, and training protocols.
- We train and evaluate this Transformer model using the simulated VRT input–output dataset, and show that it can accurately predict the VRT calculation results.

The remainder of the paper is organized as follows. Section II describes the data representation and methodology. Section III presents experimental results. Section IV concludes the paper and discusses future work.

II. METHODOLOGY

A. Data Description and Sliding-Window Representation

1) *Data Description*: In the MAPP framework, the retrieval of aerosol and ocean parameters is modeled to minimize the following cost function [3]:

$$\chi^2(\mathbf{x}) = \Phi(\mathbf{x})_{\text{data}} + \Phi(\mathbf{x})_{\text{prior}} \\ = \frac{1}{2}[(\mathbf{f} - \mathbf{y})^T \mathbf{S}_e^{-1}(\mathbf{f} - \mathbf{y}) + (\mathbf{x} - \mathbf{x}_a)^T \mathbf{S}_a^{-1}(\mathbf{x} - \mathbf{x}_a)], \quad (1)$$

where $\mathbf{f}$ represents VRT calculations that map the state vector into polarimeter measurements, i.e., $\hat{\mathbf{y}} = \mathbf{f}(\mathbf{x})$; $\mathbf{x}_a$ is the *a priori* state vector; $\mathbf{S}_e$ denotes the measurement error covariance matrix; $\mathbf{S}_a$ is the *a priori* covariance matrix that quantifies uncertainty in the initial state estimate [6]. In this work, we replace the computationally intensive forward model $\mathbf{f}$ with a Transformer model.

The dataset used in this study is generated using the MAPP simulation framework and consists of simulated VRT input–output pairs. For each observation index t the input to the Transformer model is a state vector $\mathbf{x}_t \in \mathbb{R}^{14}$, which includes 11 aerosol and ocean parameters together with 3 solar–instrument observation geometry variables. The output of the VRT model, and thus the learning target, $\hat{\mathbf{y}}_t \in \mathbb{R}^{22}$, is a 22-dimensional vector of simulated polarimeter measurements,

2) *Sliding-Window Representation*: To construct fixed-length inputs suitable for sequence modeling, we apply a sliding window to combine input and output vectors. Windows are indexed by i . For the window length T , the i -th input window is defined as

$$\mathbf{X}_i = \begin{bmatrix} \mathbf{x}_i^T \\ \mathbf{x}_{i+1}^T \\ \vdots \\ \mathbf{x}_{i+T-1}^T \end{bmatrix} \in \mathbb{R}^{T \times 14}, \quad i = 1, 2, \dots, L - T + 1. \quad (2)$$

A unit stride is used between two consecutive windows, yielding $L - T + 1$ overlapping windows.

Each window $\mathbf{X}_i$ is paired with a single retrieval target associated with the final observation in the window:

$$\hat{\mathbf{y}}_i = \mathbf{y}_{i+T-1} \in \mathbb{R}^{22}. \quad (3)$$

This formulation allows the model to use the T consecutive observations in $\mathbf{X}_i$ to predict the retrieval output at the last observation of the window as illustrated in Fig. 2. The same sliding-window representation is used for all models considered in this study.

B. The Proposed Transformer-based Model

The proposed model uses an encoder-decoder Transformer architecture following the general design of [11]. The encoder processes the entire input window to extract a sequence of contextual representations, and the decoder uses the previous step targets and the encoder output to produce a single prediction. The overall architecture is summarized in Fig. 3.

1) *Input Projection and Positional Encoding*: Each input vector $\mathbf{x}_{i,k} \in \mathbb{R}^{14}$, corresponding to the k -th position within window i , is first mapped into the d_{model} -dimensional embedding space using a linear projection

$$\mathbf{z}_{i,k} = W_{\text{enc}} \mathbf{x}_{i,k} + \mathbf{b}_{\text{enc}}, \quad (4)$$

where $W_{\text{enc}} \in \mathbb{R}^{d_{\text{model}} \times 14}$ and $\mathbf{b}_{\text{enc}} \in \mathbb{R}^{d_{\text{model}}}$. Following standard Transformer practice [11], the projected embeddings are scaled by $\sqrt{d_{\text{model}}}$ and combined with a sinusoidal positional encoding $p_k \in \mathbb{R}^{d_{\text{model}}}$:

$$\mathbf{h}_{i,k} = \sqrt{d_{\text{model}}} \mathbf{z}_{i,k} + \mathbf{p}_k, \quad (5)$$

The resulting embedded sequence $\mathbf{H}_i = [\mathbf{h}_{i,1}, \dots, \mathbf{h}_{i,T}] \in \mathbb{R}^{T \times d_{\text{model}}}$ is used as the input to the encoder.

The decoder inputs are embedded using the same procedure. Specifically, the decoder is driven by a teacher-forced target sequence shifted by one position. For window i , we define the decoder input sequence $\mathbf{Y}_i^{\text{in}} = [\mathbf{y}_i, \mathbf{y}_{i+1}, \dots, \mathbf{y}_{i+T-2}] \in \mathbb{R}^{(T-1) \times 22}$. This sequence is projected and combined with positional encoding in the same way as the encoder inputs to form $\mathbf{S}_i = [\mathbf{s}_{i,1}, \dots, \mathbf{s}_{i,T}] \in \mathbb{R}^{(T-1) \times d_{\text{model}}}$, which serves as the input to the decoder.

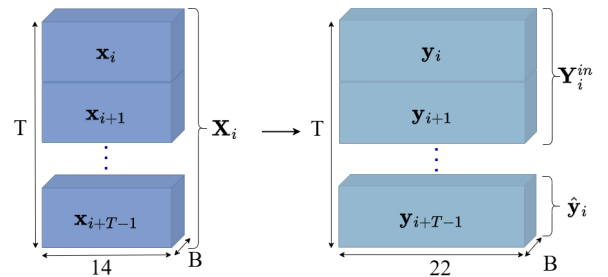


Fig. 2. Sliding-window construction for the polarimeter measurements. Consecutive observations $\mathbf{x}_t$ are grouped into overlapping windows $\mathbf{X}_i$ of length T . A training batch (batch size B) windows to form an input tensor of shape $(B, T, 14)$. Each window is paired with a target associated with the final observation in the window, $\mathbf{y}_{i+T-1}$. The shape of the output tensor is $(B, 1, 22)$.

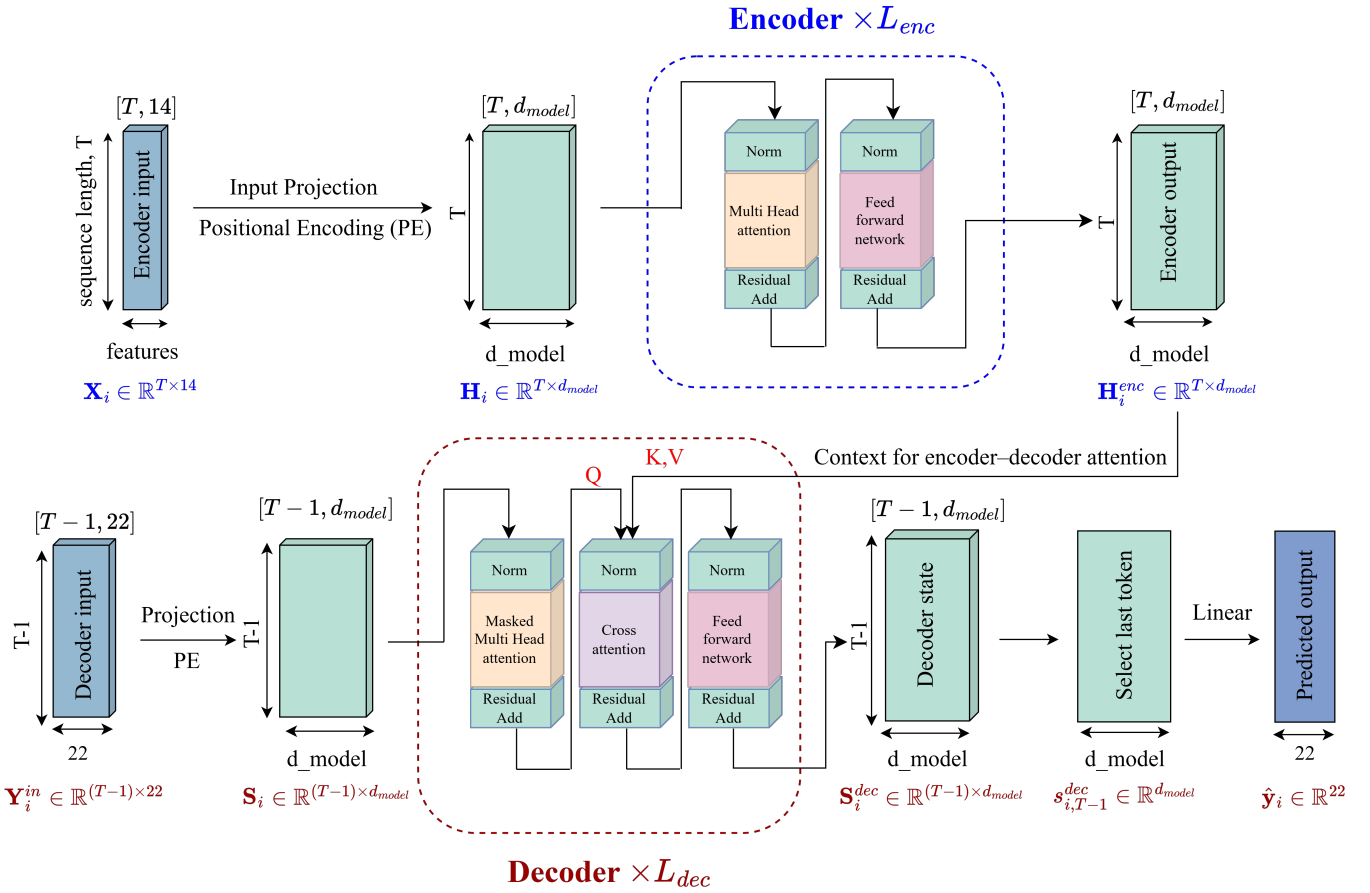


Fig. 3. Encoder-decoder Transformer architecture used in this work (shown for a single input window). Each input window $\mathbf{X}_i \in \mathbb{R}^{T \times 14}$ is projected to d_{model} and added to sinusoidal positional encodings before entering the encoder. The decoder receives a shifted target sequence $\mathbf{Y}_i^{in} \in \mathbb{R}^{(T-1) \times 22}$ during training and applies (i) masked self-attention with a causal mask, (ii) encoder-decoder cross-attention over $\mathbf{H}_i^{enc}$, and (iii) a feed-forward network, with residual connections and layer normalization around each block. The final prediction $\hat{\mathbf{y}}_i \in \mathbb{R}^{22}$ is obtained by mapping the last decoder hidden state through a linear layer. The batch dimension is omitted for clarity.

2) *Encoder Architecture*: The encoder is a stack of L_{enc} layers operating on the embedded input sequence $\mathbf{H}_i$. All attention and feed-forward sublayers adopt a pre-normalization scheme. The z-score normalization is adopted in this process. This pre-norm design improves optimization stability, especially for deeper Transformer stacks [12]. A final layer normalization is applied at the encoder output.

Multi-head self-attention is calculated as the scaled dot-product, following the standard formulation in [11]. The feed-forward network (FFN) consists of two linear transformations, with a nonlinear ReLU activation function in between. The dimension change in the FFN is: $d_{model} \rightarrow d_{ff} \rightarrow d_{model}$, where d_{ff} is the dimension of the inner-layer. The FFN is applied to the multi-head attention output [13].

3) *Decoder Architecture*: The decoder is composed of L_{dec} stacked layers. Each decoder layer contains three sublayers: (i) masked multi-head self-attention, (ii) encoder-decoder cross-attention, and (iii) FFN. During training, the decoder self-attention sublayer uses a causal mask to ensure that each position can only attend to current and past target positions, preventing access to future information. The same pre-

normalization and residual connection scheme used in the encoder is applied to all decoder sublayers. A final layer normalization is applied at the output of the decoder stack.

Let $\mathbf{S}_i^{dec} = [\mathbf{s}_{i,1}, \dots, \mathbf{s}_{i,T-1}] \in \mathbb{R}^{(T-1) \times d_{model}}$ denote the output of the decoder stack for the i -th sliding window after the final layer normalization. The hidden state at the last decoder position, $\mathbf{s}_{i,T-1}$, is used as a summary representation of the window and mapped to the output space using a learned linear projection:

$$\hat{\mathbf{y}}_i = \mathbf{W}_{out} \mathbf{s}_{i,T-1} + \mathbf{b}_{out}, \quad \mathbf{W}_{out} \in \mathbb{R}^{22 \times d_{model}}, \quad (6)$$

where $\hat{\mathbf{y}}_i \in \mathbb{R}^{22}$ denotes the estimated target for the i -th sliding window.

4) *Loss Function and Model Parameters*: The model is trained end-to-end using the loss function of mean squared error (MSE) [14]:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{\mathbf{y}}_i - \mathbf{y}_{i+T-1})^2, \quad (7)$$

where N denotes the total number of sliding windows in the dataset.

TABLE I
MSE OF TRANSFORMER VARIANTS FOR DIFFERENT MODEL SIZES AND WINDOW SIZES.

Params (M)	$T = 3$	$T = 10$	$T = 100$
0.9	5.015×10^{-6}	1.422×10^{-5}	1.220×10^{-5}
1.4	8.070×10^{-6}	2.851×10^{-6}	2.988×10^{-5}
2.7	2.570×10^{-6}	3.881×10^{-6}	1.771×10^{-5}
4.0	4.305×10^{-6}	8.378×10^{-6}	1.163×10^{-5}

TABLE II
MSE OF CNN, RNN, AND TRANSFORMER MODELS FOR DIFFERENT WINDOW SIZES.

Model	Window size, T	MSE $\downarrow$	Total MSE $\downarrow$
CNN	3	9.742×10^{-6}	0.0076
CNN	10	1.383×10^{-5}	0.0108
CNN	100	2.962×10^{-5}	0.0231
RNN	3	1.436×10^{-6}	0.0011
RNN	10	2.639×10^{-6}	0.0021
RNN	100	1.819×10^{-6}	0.0014
Transformer	3	2.570×10^{-6}	0.0020
Transformer	10	2.851×10^{-6}	0.0022
Transformer	100	1.163×10^{-5}	0.0091

We evaluate several Transformer configurations by varying the model dimension d_{model} and the number of encoder and decoder layers ($L_{\text{enc}}, L_{\text{dec}}$), while keeping the number of attention heads H and the hidden dimension of the FFN inner-layer d_{ff} fixed. In the experiments, we use $H = 4$ heads and $d_{\text{ff}} = 512$, and consider $d_{\text{model}} \in \{128, 256\}$ with $(L_{\text{enc}}, L_{\text{dec}}) \in \{(2, 2), (3, 3)\}$, resulting in approximately 0.9M, 1.4M, 2.7M, and 4.0M trainable parameters.

C. Training Details

For the Transformer experiments, we use the Adam optimizer [15] with an initial learning rate of 5×10^{-4} , and a batch size of 256. A learning-rate scheduler is employed, which decreases the learning rate by a factor of 0.5 when the validation loss does not improve for a specified patience period. To stabilize optimization, gradient norms are clipped to a maximum value of 1.0. Dropout is applied for regularization, with a value of 0.1 for the largest Transformer configuration and is adjusted within the range of 0.1–0.2 for smaller variants.

Besides the proposed Transformer model, we also trained CNN and RNN models for comparing the prediction results. The CNN and RNN models are retrained under the same optimizer and loss function.

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments are conducted using data generated by the MAPP framework described in Section II. The dataset consists of aerosol and ocean parameters, sun-instrument geometry

parameters, and the corresponding polarimeter measurements produced by VRT calculations [3]. Sliding-window inputs are constructed with three window lengths, $T \in \{3, 10, 100\}$, and the same windowed representation is used for all models, including CNN, RNN, and Transformer architectures.

All models are trained using the MSE loss between the predicted retrieval and the VRT-based reference values. In addition to the average test MSE, we also report the total MSE, defined as the sum of MSEs over all windows. For a given model, the Total MSE can be found via $\text{Total MSE} = N_{\text{test}} \times \text{MSE}$, where N_{test} denotes the number of sliding windows in the testing dataset.

All experiments are executed on a workstation equipped with an Intel(R) Core(TM) Ultra 7 265 CPU (2.40 GHz) and an NVIDIA RTX 2000 Ada Generation GPU. GPU acceleration is enabled via PyTorch 2.5.1 [16] with CUDA 12.1 and cuDNN 9.1.

B. Model Selection and Comparison with CNN and RNN

All models are evaluated using the same testing dataset, which contains 200,000 input–output pairs. Table I reports the MSE of the proposed Transformer under different window sizes (3, 10, and 100) and model sizes (0.9M, 1.4M, 2.7M, and 4M parameters). For a fixed window size, increasing model size does not always lead to improved prediction accuracy. Specifically, when the window size is 3, the 2.7M-parameter model achieves the lowest MSE; when the window size is 10, the best performance is obtained with the 1.4M-parameter model; when the window size is 100, the lowest MSE is achieved by the 4M-parameter model.

Based on these results, the best-performing Transformer models for each window size are selected and compared with CNN and RNN models. The comparison results are summarized in Table II. Across all model types, a window size of 3 yields the best prediction performance.

The relative performance between the Transformer and RNN varies with window length. For $T = 3$, the RNN achieves lower MSE than the Transformer; for $T = 10$, the Transformer and RNN achieve nearly the same MSE; for $T = 100$, the Transformer shows higher error than the RNN. This result suggests that a moderate window length allows the Transformer to make effective use of the information without unnecessary complexity.

C. Parity Plots on Prediction v.s. Ground Truth

Fig. 4 shows parity plots for the best performing Transformer configuration ($T = 3$ and the parameter size is 2.7M) on the testing dataset. In each output channel, the horizontal axis shows the ground truth values and the vertical axis shows the corresponding Transformer prediction. Both axes are normalized to 0 to 1 in the plot.

It can be observed that the samples cluster tightly around the one-to-one line, indicating that the predicted values closely match the ground truth. To quantify this agreement, we report the coefficient of determination R^2 [14], which measures how well the predictions explain the variability in the reference

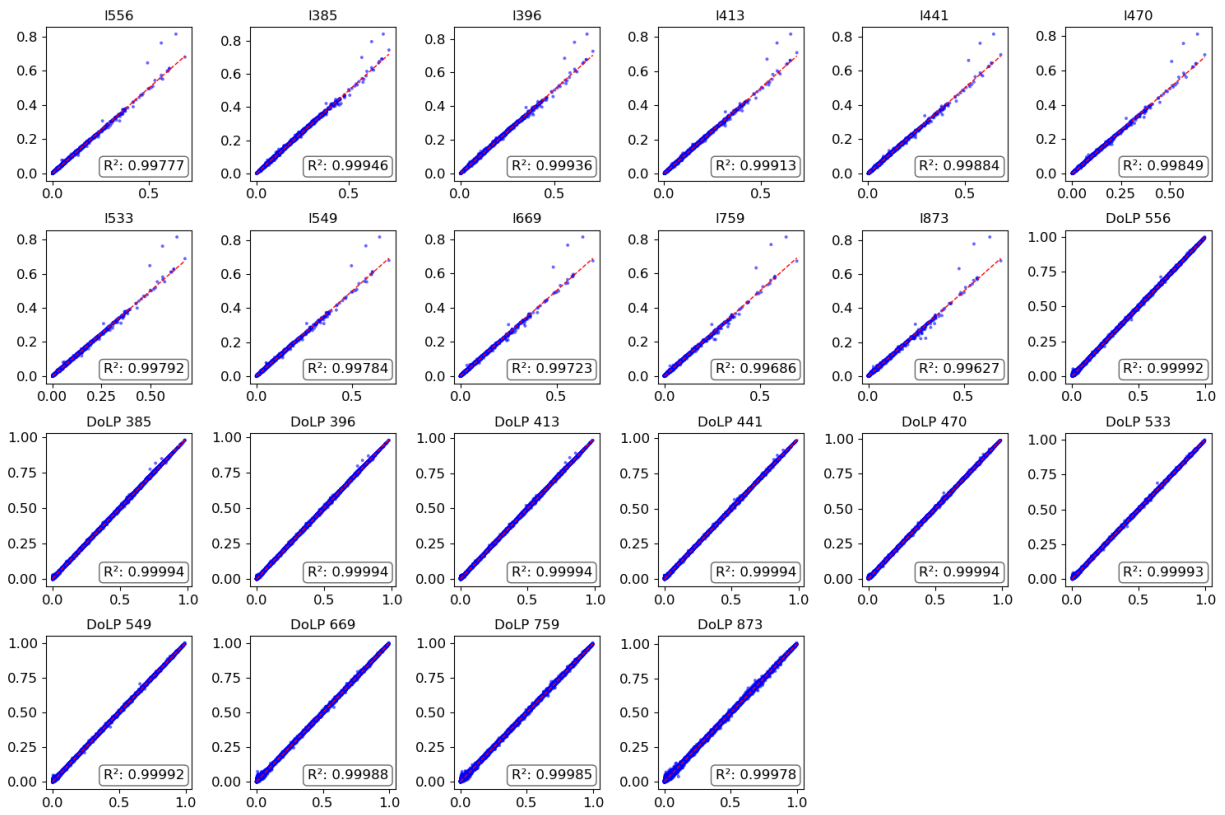


Fig. 4. Parity plots for the best-performing Transformer model on the testing dataset. Each plot compares VRT-based reference values (horizontal axis) with Transformer predictions (vertical axis) for radiance I (first 11 plots) and degree of linear polarization (DoLP; later 11 plots). The red dashed line denotes the one-to-one relationship. The coefficients of determination (R^2) are near unity across channels (typically > 0.997), indicating close agreement between the Transformer predictions and VRT calculation results.

values: R^2 values closer to 1 indicate stronger agreement. Across all output channels, (R^2) are consistently high (> 0.997), suggesting minimal bias and high predictive accuracy. These visual results support the quantitative findings in Tables I, showing that the Transformer model produces accurate predictions that match the VRT calculation results.

IV. CONCLUSION

This paper presents a Transformer-based deep-learning approach to replace the computationally intensive VRT calculations used in the MAPP algorithm. The proposed Transformer operates on sliding windows of consecutive input vectors and predicts the polarimeter measurement vector associated with the final element of each window. Under a unified training and evaluation protocol, the Transformer achieves the MSE on the order of $10^{-6} - 10^{-5}$ on the testing dataset. The results show that the Transformer consistently yields lower error than the CNN model and performance comparable to the RNN model, while maintaining stable accuracy as the context increases. Parity plots further confirm close agreement with the VRT-based reference measurements. Future work will extend this approach to longer contexts, investigate alternative attention mechanisms, and assess performance on real PACE observations.

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